Optimal infectious-vaccination containment costs on infinite Cartesian and strong grids

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Editorial note. One combined infinite-grid containment object, including Cartesian and strong metrics, exact inoculation costs and constructive time bounds. Complete original mathematical paper131–1336, all native TikZ figures and embedded bibliography retained. The only native TeX input tikz_stuff is expanded verbatim as a preamble span; no active TeX input remains.

Editorial note. Finite-graph algorithms/parameterized results, individual dimensions, illustrative examples and direct hypercube-strength tabular consequences are not additional counts. Original process conventions, schedules and proofs retained without modification.

Editorial note. Authors retain responsibility for mathematics. No proof repair, new hypotheses, independent proof review, process simulation, strategy/counterexample search or certification.

Editorial note. Actual r001 missing tikzscale.sty: no graphics scaling or standalone calls occur; omit unused tikzscale/standalone imports only in rendering. Original native TikZ helpers/figures and mathematical text unchanged.

Editorial note. Actual nullfont right-parenthesis from redundant parentheses in native TikZ shift at source line613: remove only the redundant pair around the same coordinate. Original graph, labels and vertex data unchanged.

Allowed standard prerequisites (editorial).

- Infinite Cartesian/strong lattice metrics, spreading-defense process, transmission sets and hypercube strength orderings. Native n>=2, root zero, unit-per-turn budget, defense/infection substep ordering and finite containment-strategy definitions retained literally. Hypercube-strength values and classical Firefighter comparisons remain original cited background.. Complete original lower obstructions and constructive containment schedules supplied with all native TikZ illustrations; no simulation, independent strategy search or proof correction.

1 Introduction

The Firefighter problem (FIREFIGHTER) is a deterministic, discrete-time game that models the spread of fire on a graph. Originally introduced by Harnell [14], FIREFIGHTER and its variants have been well-studied; see the survey by Finbow and MacGillivray [9].

In a game of FIREFIGHTER, vertices can be in one of three states: *open, burning* or *defended.* All vertices are initially open. However, once burning or defended, a vertex remains so for the rest of the game. An instance of FIREFIGHTER on a rooted graph G proceeds as follows. At time-step $t = 0$, the fire breaks out at root r. For each time-step $t > 0$, the firefighter selects a vertex to defend, and then the fire spreads to the open neighbours of all burning vertices. If there is some time-step at which no open vertex is adjacent to a burning vertex, we declare the fire to be *contained* and count the number of *saved* vertices, those either defended or open and therefore unreachable by the fire. The number of time-steps from outbreak to containment of a particular instance is the *lifetime* of the instance. A *certificate* for an instance of FIREFIGHTER is a *strategy*, a sequence of vertices selected for defence (in order of when they are defended).

In the generalised model, b-FIREFIGHTER, up to b vertices are defended at each time-step for some fixed budget b. We consider a variant of FIREFIGHTER and b-FIREFIGHTER called the Infectious Vaccination problem (INFECTVAX). As INFECTVAX is inspired by studies of spreadable –or self-disseminating– animal vaccines in combatting viruses, we refer to the fire as the *contagion* that spreads during each time-step. All defended vertices are called *vaccinated* and those that are defended directly are called *inoculated.* INFECTVAX proceeds as b-FIREFIGHTER with an inoculation budget b per time-step, but the vaccination spreads in the same way as the contagion. That is, for INFECTVAX: all vertices are initially open and at time-step 0, the contagion breaks out at a root r of graph G. During each time-step $t > 0$,

1. the vaccine spreads from all vaccinated vertices to their open neighbours;
2. up to b vertices are inoculated, for some integer budget b; and
3. the contagion spreads to the open neighbours of all contagious vertices.

An instance of INFECTVAX is composed of a graph G, a root $r \in V(G)$, a budget $b \in \mathbb{N}$ and a target number of vertices to save $k \in \mathbb{N}$. We represent an instance of INFECTVAX as the tuple (G, r, b, k). Interestingly, INFECTVAX has been studied as the *spreading vaccination model* by Anshelevich et al. [1] and Floderus et al. [10]. Both studied it as an optimisation problem with two objectives: maximising the number of saved vertices and minimising the budget used. Anshelevich et al. [1] prove the existence of constant-factor approximation algorithms for maximising saved vertices, and use work by Floderus et al. [10] to show that there is no $((1 - o(1)) \ln n)$ approximation (unless $NP \subseteq DTIME(n^{\mathtt{poly}\ln(n)})$) for minimising the budget, but that there is a polynomial-time $\ln n$-factor approximation algorithm for this objective [1]. They observe that INFECTVAX is easier to approximate than FIREFIGHTER.

We begin with terminology and formal definitions in Section 2. In Section 3, we show that for an interval graph G on n vertices and m edges, INFECTVAX on input (G, r, b, k) can be solved in $\mathcal{O}(n + m)$ time. In Section 4, we consider a parameterised approach and show that INFECTVAX is FPT (fixed-parameter tractable) when parameterised by budget and vertex cover size, or by budget and neighbourhood diversity. We further show in Section 4.3, that INFECTVAX is in XP when parameterised by maximum path length. With budget $b = 1$, we study two vertex-transitive families of infinite graphs in Section 5: the infinite n-dimensional Cartesian grid and the infinite n-dimensional strong grid. We provide strategies which guarantee containment on both graph families. Since the number of vertices saved is infinite, we bound the number of time-steps until containment occurs, which in turn bounds the number of contagious vertices. Finally, in Section 5.3, we show that for a fixed budget b the number of inoculations required to contain the virus is not closed under induced subgraphs.

2 Preliminaries

We begin with essential graph theoretic definitions and recommend [20] for a more complete overview. We assume all graphs to be undirected and unweighted. An *induced subgraph* $H = (V_H, E_H)$ of $G = (V_G, E_G)$, denoted $G[V_H]$, is a subgraph of G on the vertices V_H such that for all $v, w \in V_H$, if $(v, w) \in E_G$, then $(v, w) \in E_H$. If the graph G is clear from the context, we use simplified notation $G = (V, E)$. For $v \in V(G)$, let $N(v)$ and $N[v]$ denote the open and closed neighbourhoods of v, respectively.

A *strategy* for a game of FIREFIGHTER is a sequence of vertices in which the ith vertex is selected for defence at the ith time-step. In INFECTVAX, we take as input an integer *budget* b that bounds the number of inoculations (direct defences) per time-step. Thus, strategies are sequences of sets of vertices. The following defines such strategies for variants such as INFECTVAX and b-FIREFIGHTER.

Definition 2.1 (Strategy with multiple defences per time-step)**.** For an instance of a FIREFIGHTER variant on a rooted graph G in which some $b \in \mathbb{N}$ vertices can be defended per time-step, a strategy is a sequence of ℓ sets of vertices

$$\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_\ell)$$

such that $\sigma_i \subseteq V(G)$ is the set of vertices defended during time-step i and $|\sigma_i| \leq b$.

The *length* of the strategy is the number of sets in the sequence.

Using this definition, b-FIREFIGHTER is formally defined as follows [3].

b-FIREFIGHTER

Instance: A rooted graph (G, r), an integer budget $b \geq 1$ and an integer $k \geq 1$.
Question: Is there a strategy σ such that if the fire breaks out at r:

- for each $\sigma_i \in \sigma$, by time i each vertex in σ_i is neither burning nor defended,
- the number of vertices defended in each time-step is at most b,
- after time $|\sigma|$, no open vertex is adjacent to a burning vertex, and
- at time $|\sigma|$, at least k vertices are saved by implementing σ?

For b-FIREFIGHTER on infinite graphs, a common consideration is whether the fire can be *contained,* which is to ask whether we can deploy defences to enclose the fire to a finite set of vertices within a finite number of time-steps. When we say the fire is *contained* on a finite graph, we require that, when the fire can no longer spread, there is at least one undefended and unburned vertex in the same component as the root vertex but inaccessible by the fire because of the defences placed. This intentionally excludes the case in which the fire cannot spread further because all undefended vertices of a finite graph have been burned. The following is attributed to Moeller and Wang [19].

Theorem 2.2. *On the infinite 2-dimensional Cartesian grid, containment is not possible for* FIREFIGHTER *but is possible for b-*FIREFIGHTER *with budget $b \geq 2$ in fewer than 8 time-steps.*

In stark contrast, we show in Section 5 that the contagion can be contained for INFECTVAX on the infinite n-dimensional Cartesian grid with a budget of $b = 1$ per time-step by the end of time-step $n^2 + 1$.

Next we formally define INFECTVAX. Recall that we view the vertices of a graph as being in one of three states at the end of each time-step: open, contagious, or vaccinated. Once a vertex is in the contagious or vaccinated state, it remains in that state forever. A vertex that is neither contagious nor vaccinated is open. A further subtlety we note is that inoculated vertices are vaccinated, but vaccinated vertices are not necessarily inoculated.

On a graph $G = (V, E)$ at time $t = 0$, the contagion breaks out at vertex $r \in V$. At each subsequent time-step, all open vertices adjacent to vaccinated vertices become vaccinated, then we choose up to b additional open vertices to inoculate, and then the contagion propagates to all open neighbours of contagious vertices. This continues until the contagion cannot spread.

INFECTVAX

Instance: A graph $G = (V, E)$, a root vertex $r \in V$, a budget $b \in \mathbb{N}$ and target $k \in \mathbb{N}$.
Question: If the contagion breaks out at r in time-step 0 and in each subsequent time-step $t > 0$:

- each open neighbour of each vaccinated vertex becomes vaccinated, then
- we select up to b open vertices to inoculate, and then
- each open neighbour of each contagious vertex becomes infected,

is there a strategy σ, such that:

- after time $|\sigma|$, no open vertex is adjacent to a contagious vertex, and
- at least k vertices are saved by time $|\sigma|$?

As an example, suppose the contagion breaks out at the origin $(0, 0)$ on an $n \times n$ Cartesian grid, where $n \geq 8$. In Figure 1, the origin is indicated by the red circled vertex labelled "0". Fix $b = 1$, so at most one vertex will be inoculated during each time-step and let $\sigma = (\{(1, 0)\}, \{(-2, 0)\}, \{(0, 3)\}, \{(0, -4)\}, \emptyset)$. The first event during each time-step is the vaccination spread: during time-step 1, there is no vaccination to spread. The second event during each time-step is that b vertices are inoculated: during time-step 1, the solid green vertex at $(1, 0)$, labelled "1" is inoculated. The third event during each time-step is the spread of the contagion: during time-step 1, the contagion spreads to the three red circled vertices labelled "1". Continuing this process, the reader can see that after the vaccine spreads during time-step 5, there is no vertex to which the contagion can spread and no need to inoculate a fifth vertex. Thus some of the σ_i may be empty.

Like FIREFIGHTER and b-FIREFIGHTER, we also consider whether containment can occur for INFECTVAX on infinite graphs. Thus, in Section 5 where containment is the objective, we omit k from consideration (since

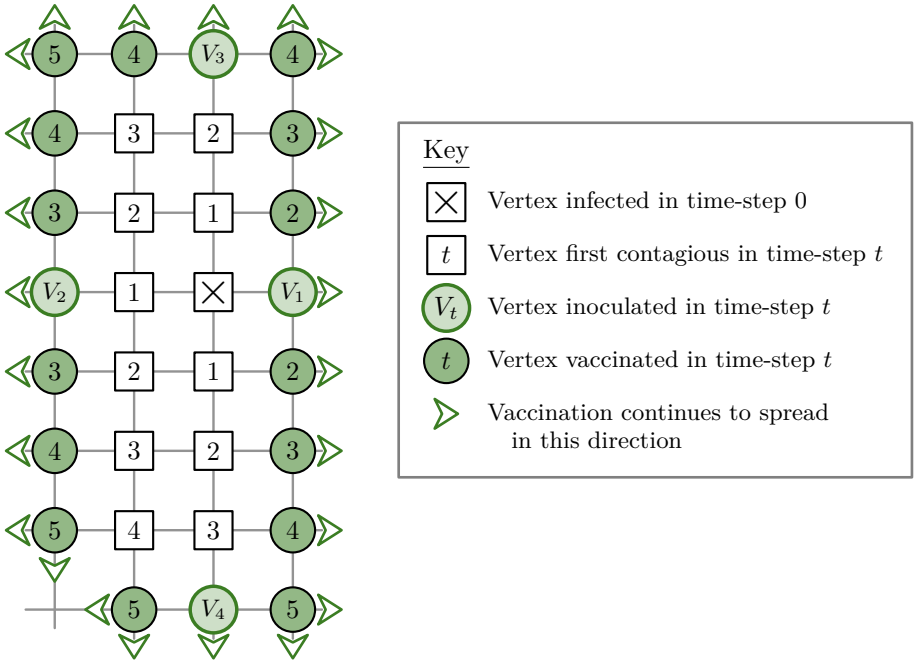


Figure 1: Using budget $b = 1$, a strategy for InfectVax on the infinite 2-dimensional Cartesian grid.

an infinite number of vertices will be saved by containment) and regard an instance of InfectVax to be the triple (G, r, b). In the case where containment occurs for a particular value of b, we will seek the minimum number of inoculations needed for containment occur.

3 Algorithmic and hardness results

In this section, we complement an existing hardness result on InfectVax with several tractability results on restrictive graph classes inspired by parallel results on more classical Firefighting problems.

Anshelevich et al. [1] observed that InfectVax is equivalent to b-Firefighter when restricted to trees. Informally, this is because a vertex in a tree is saved if it has at least one vaccinated (defended) ancestor, so the spread of the vaccine does not change the number of saved (vaccinated) vertices. Thus, existing results for b-Firefighter on trees follow immediately for InfectVax on trees, including NP-completeness.

Theorem 3.1 ([1])**.** InfectVax *is NP-complete.*

A tractability result and an approximation result immediately following from Bazgan et al. [3] and Cai et al. [6] respectively:

Theorem 3.2. InfectVax *with budget b is solvable in time polynomial in input tree size on trees of maximum degree $b + 2$, rooted at a vertex of degree $b + 1$.*

Theorem 3.3. *There is a $(1 - 1/e)$-approximation algorithm for* InfectVax *on trees.*

We next consider interval graphs. Let $G = (V, E)$ be a graph on n vertices such that there exists a set $I = \{I_1, \ldots, I_n\}$ of intervals on the real line and a bijection $\mathcal{I} : V \to I$ that maps each vertex $v \in V$ to an interval $\mathcal{I}(v) = I_v$ such that, for all $u \in N(v)$, $I_u \cap I_v \neq \emptyset$ and for all $w \notin N(v)$, $I_v \cap I_w = \emptyset$. Such a pair $(I, \mathcal{I})$ is called an *interval representation* and a graph for which there exists an interval representation is called an *interval graph*. Given an interval graph $G = (V, E)$, we can obtain an interval representation in $O(|V| + |E|)$ time [5].

Adapting an approach of Fomin et al. [11], we show that in any instance of InfectVax on an interval graph, the contagion can be contained within at most three time-steps and we can resolve InfectVax in

polynomial time on interval graphs. We essentially do this by first playing optimally on one side of the contagion, and then on the other. As in [11], we assume without loss of generality that interval endpoints are unique, since intervals can be perturbed slightly if necessary.

Theorem 3.4. *Given an instance (G, r, b, k) of* InfectVax *on an interval graph G with n vertices and m edges, this instance can be solved in $\mathcal{O}(n+m)$ time.*

Proof. First, if $b \geq N(r)$ then we inoculate the neighbourhood of r and immediately restrict the contagion to a single vertex.

Second, if $N(r) \subseteq N[v]$ for some vertex v then during time-step 1, we inoculate v along with up to $b-1$ neighbours of r. This restricts the contagion to $\max(1, |N[r]|-b)$ vertices if $v \in N(r)$, and $\max(1, |N[r]|-b+1)$ otherwise; both are best possible.

Otherwise, we observe that there can be no single neighbour of r whose interval extends both farther to the left and farther to the right than the interval of any other neighbour of r. Let v_ℓ and v_r be the unique neighbours of r such that the left endpoint of $\mathcal{I}(v_\ell)$ extends farther to the left than the interval of any other neighbour of r and the right endpoint of $\mathcal{I}(v_r)$ extends farther to the right than the interval of any other neighbour of r.

If $b \geq 2$ then we can solve the instance optimally by inoculating both v_ℓ and v_r as well as up to $b-2$ other distinct members of $N(r)$ in time-step 1, therefore restricting the contagion to $\max(1, |N[r]| - b)$ vertices.

The remaining case is that $b = 1$ and v_ℓ and v_r are distinct. We argue that we can contain the contagion in three time-steps, in a way that is a best-possible solution. For this, we need to refer to the neighbours of neighbours of r that lie to the left of r as separate from those on the right. We therefore denote by $L_2(r)$ the set of vertices that are distance two from r and correspond to intervals that lie entirely to the left of $I(r)$ in $\mathcal{I}$. And symmetrically, we denote by $R_2(r)$ the set of vertices that are distance two from r and correspond to intervals that lie entirely to the right of $I(r)$ in $\mathcal{I}$.

For $b = 1$, we next argue that if v_ℓ and v_r are distinct then under any inoculation strategy at least $|N(r)| + \min(|L_2(r)|, |R_2(r)|) - 1$ vertices will be infected, and we then argue that we can find a strategy that achieves this in the required time complexity. We consider two cases, depending on whether a vertex in $N(r)$ is inoculated in time-step 1 or not.

If not, then $|N(r)|$ vertices become infected in time-step 1. Because the vertex inoculated in time-step 1 was not a neighbour of r, its interval must lie entirely to the right or entirely to the left of $\mathcal{I}(r)$: without loss of generality, entirely to the right. Observe that when the vaccine spreads at the beginning of time-step 2, it cannot spread to any member of $L_2(r)$. During the inoculation-selection phase in time-step 2, at most one member of $L_2(r)$ is inoculated, and then at the end of that time-step all other members will become infected. Therefore at least $|N(r)| + |L_2(r)| - 1 + |\{r\}| = |N(r)| + |L_2(r)|$ vertices are infected.

Suppose we inoculate within $N(r)$ during time-step 1. Then at most one of v_r, v_ℓ is inoculated in time-step 1 and $|N(r)| - 1$ vertices become infected in time-step 1. When the vaccine spreads during time-step 2, at most one of $L_2(r)$ or $R_2(r)$ become vaccinated: without loss of generality, suppose the vaccination does not spread to $L_2(r)$ during time-step 2. Then no neighbour of a vertex of $L_2(r)$ was inoculated on time-step 1. During the inoculation-selection phase in time-step 2, at most one vertex of $L_2(r)$ can be inoculated and then at the end of that time-step all other vertices of $L_2(r)$ will become infected. At least $|N(r)| + |L_2(r)| - 1$ vertices are infected by the end of time-step 2.

Thus, at least $|N(r)| + \min(|L_2(r)|, |R_2(r)|) - 1$ will be infected. This bound can be achieved by the following inoculation strategy:

- in time-step 1: inoculate v_ℓ if $|L_2(r)| > |R_2(r)|$, else inoculate v_r;
- in time-step 2: inoculate the vertex of $L_2(r)$ with the leftmost left endpoint if we previously inoculated v_r, else inoculate the vertex of $R_2(r)$ with the rightmost right endpoint.

With the above strategy, the contagion is contained after the vaccine spreads during time-step 3, with $|N(r)| + \min(|L_2(r)|, |R_2(r)|) - 1$ vertices infected. Determining which of $L_2(r)$ or $R_2(r)$ is larger, and identifying the vertices with extremal intervals that we require can be achieved from an adjacency list and the interval representation in $\mathcal{O}(|V| + |E|)$ time. ■

4 Parameterised approaches

If a problem A parameterised by k can be solved in $\mathcal{O}(n^{f(k)})$ time for input size n and a computable function f, we say A is in XP, meaning *slicewise polynomial* [8]. More strongly, A is *fixed-parameter tractable* (FPT) with respect to k if it can be solved in $\mathcal{O}(f(k)n^{\mathcal{O}(1)})$ time. We show that InfectVax is FPT when parameterised by the combination of budget with vertex cover size (Section 4.1) or neighbourhood diversity (Section 4.2). It is in XP when parameterised by the combination of budget and maximum path length (Section 4.3).

Lemma 4.1. *Let $\mathcal{P}$ be a parameterised decision problem on a graph $G = (V, E)$ on n vertices for which a certificate is a finite sequence of subsets of vertices $\sigma = (\sigma_t)_{t \leq T}$ for some $T \in \mathbb{N}$. Let f, g and h be computable functions and suppose:*

1. *The lifetime of any instance is bounded by $f(k)$ (that is, $T \leq f(k)$).*
2. *For each time-step t, V can be partitioned into at most $g(k)$ equivalence classes such that selecting any two vertices from the same class for σ_t yields identical outcomes for the rest of the process.*
3. *The outcome of a fixed certificate σ can be simulated in $h(n, k)$ time.*

Then $\mathcal{P}$ can be solved in $\big(g(k)\big)^{f(k)} \cdot h(n, k)$ time. In particular, if $\mathcal{P}$ can be expressed as $F(k) \cdot \mathrm{poly}(n)$ for some computable F, then $\mathcal{P}$ is fixed-parameter tractable.

Proof. By 1., any strategy has length at most $f(k)$. By 2., there are at most $g(k)$ outcome-distinct choices per step, hence at most $g(k)^{f(k)}$ candidate certificates overall. By 3., simulating a fixed candidate takes $h(n, k)$ time. Hence the total runtime is $g(k)^{f(k)} \cdot h(n, k)$. ■

Two vertices u, v in graph G have the same *neighbourhood type* if $N(u) \backslash \{v\} = N(v) \backslash \{u\}$. A *type partition* of G is a partition $\mathcal{T} = \{T_1, \ldots, T_w\}$ of $V(G)$ such that if u and v have the same neighbourhood type, then $u, v \in T_i$, for some $i : 1 \leq i \leq w$. This is an equivalence relation, whose classes are the types. Suppose that two strategies σ and σ' are identical except during some time-step t, $u \in T_i$ is inoculated via σ, whereas $v \in T_i$ via σ'. If the entire evolution is identical up to a permutation of vertices within T_i, then we say the strategies produce *equivalent outcomes*. That is, if the final partition into contagious, vaccinated, and open states is the same up to relabelling within T_i, then σ and σ' produce equivalent outcomes.

Lemma 4.2. *Let G be a graph that admits a type partition $\mathcal{T} = \{T_1, \ldots, T_w\}$. In an instance of* InfectVax *on G, inoculating different vertices from the same type yields equivalent outcomes.*

Proof. Let $G = (V, E)$ be a graph with a type partition $\mathcal{T} = \{T_1, \ldots, T_w\}$. Let σ and σ' be two strategies that are identical for the first t time-steps, with the exception that during time-step t, open vertex $u \in T_i$ is inoculated by σ and open vertex $v \in T_i$ is inoculated by σ'. After $t-1$ time-steps, the state configurations are identical; that is, regardless of whether one follows strategy σ or σ', at the end of time-step $t-1$, vertex states are the same. By definition, $G[T_i]$ is either a clique or an independent set. Consequently, during time-step t, the spread of the vaccine and the contagion to vertices outside T_i occurs identically, regardless of whether u or v is inoculated at time-step t. In T_i, all vertices other than u and v have identical neighbourhoods and identical states at the start of the time-step, so the spread of the vaccine or contagion within T_i is also identical under both strategies. Hence, at the end of time-step t, the configurations remain identical up to permutation of vertices within T_i. Consequently, inoculating different vertices from the same type produces equivalent outcomes. ■

4.1 Vertex cover

A *vertex cover* of a graph $G = (V, E)$ is a set $S \subseteq V$ such that every edge of G has at least one endpoint in S. The minimum size of such a set is the *vertex cover number* s of G. For an instance (G, r, b, k) of InfectVax where $G = (V, E)$ has a vertex cover S of size s, let $\mathcal{C} = \{C_1, C_2, \ldots, C_{2^s}\}$ be the collection of all possible subsets of S that could appear as neighbourhoods of vertices in $V \setminus S$. Define $\chi_{G,S} : V \setminus S \to \mathcal{C}$ by $\chi_{G,S}(v) = N(v) \cap S$ for $v \in V \setminus S$. The equivalence classes on $V \setminus S$ induced by $\chi_{G,S}$ are called *$\chi_{G,S}$-classes*. There are at most 2^s such classes among vertices in $V \setminus S$.

Lemma 4.3. *The lifetime of an instance of* InfectVax *on a graph with vertex cover number s is at most $2s + 2$.*

Proof. Let S be a vertex cover of size s. Note that $V \backslash S$ is an independent set. At worst, S is also an independent set, and the graph is thus bipartite. Regardless of the part in which the contagion originates, since the graph is bipartite, infection must spread back and forth between S and $V \backslash S$. There is a maximum of s time steps to infect all of S and, at worst, it takes s time steps to infect all of $V \backslash S$. Thus, accounting for the initial outbreak at time-step 0 and a final step to certify containment yields the bound $2s + 2$ on the lifetime. ■

Theorem 4.4. *Given an instance (G, r, b, k) of* InfectVax *on a graph G with n vertices, m edges, and vertex cover of size s, this instance can be solved in $2^{\mathcal{O}(s^2 b)} \cdot (n + m)$ time. Thus,* InfectVax *is FPT by vertex cover number and budget.*

Proof. Observe that the $\chi_{G,S}$-classes, taken together with S, forms a type partition of G, with size bounded by $2^s + s$. By Lemma 4.2, we need only consider strategies that specify, for each time-step, which types to inoculate rather than specific vertices. At each time-step, we inoculate at most b vertices. For each inoculated vertex, we choose amongst the at most $2^s + s$ types. The number of outcome-distinct ways to choose at most b vertices per time-step is at most $2^{\mathcal{O}(sb)}$: for $b > 1$ we have this from $\sum_{i=0}^{b} \binom{s+2^s}{i} \leq (s + 2^s)^b = 2^{\mathcal{O}(sb)}$, for $b = 1$ we have this from $\sum_{i=0}^{b} \binom{s+2^s}{i} = 1 + s + 2^s = 2^{\mathcal{O}(sb)}$.

By Lemma 4.3, we have that $f(s) = 2s + 2$. There are $g(s) = 2^{\mathcal{O}(sb)}$ strategy choices per time-step and a simulation time of $h(n, s) = O(n + m)$. By Lemma 4.1, we obtain an algorithm running in $(2^{\mathcal{O}(sb)})^{2s+2} \cdot O(n+m) = 2^{O(s^2 b)} \cdot O(n+m)$ time, so InfectVax is FPT when parameterised by vertex cover number and budget. ■

4.2 Neighbourhood diversity

Lampis [18] introduced a generalisation of vertex cover called *neighbourhood diversity*. Recall that two distinct vertices u, v are of the same *neighbourhood type* if $N(u) \backslash \{v\} = N(v) \backslash \{u\}$. The *neighbourhood diversity* of a graph G is the minimum number w of sets in a partition $T_1, T_2, \ldots, T_w$ of $V(G)$ such that every vertex in T_i has the same neighbourhood type for $i \in \{1, 2, \ldots, w\}$. A *type partition* or a *neighbourhood partition* of G is a partition $\mathcal{T} = \{T_1, T_2, \ldots, T_w\}$. By definition, for any type T_i, the induced subgraph $G[T_i]$ is either a clique or an independent set, and for any two distinct types T_i, T_j, either the vertices in T_i are adjacent to every vertex in T_j or none.

Lemma 4.5. *The lifetime of an instance of* InfectVax *on a graph with neighbourhood diversity w is at most w.*

Proof. Let $\mathcal{T} = \{T_1, T_2, \ldots, T_w\}$ be a type partition of G. Define the *type graph* H as the graph with vertex set $\mathcal{T}$, where two vertices T_i and T_j are adjacent in H if and only if in G every vertex in T_i is adjacent to every vertex in T_j.

Initially, only the type containing the root has an infected vertex. At each subsequent time-step, if the process has not terminated, there exists an open vertex adjacent to an infected vertex; thus there is a type T_j that is adjacent in H to a type T_i containing an infected vertex. By definition of H, every open vertex of T_j has an infected neighbour in T_i, so when the contagion spreads, all open vertices of T_j become infected. If a type T contains an infected vertex and $G[T]$ is a clique, all open vertices of T become infected in the same substep. If instead $G[T]$ is an independent set, there is no internal spread. Hence, in each time-step the number of types that contain at least one infected vertex increases by at least one. Since there are only w types in total, the process can last at most w time-steps. ■

Theorem 4.6. *Given an instance (G, r, b, k) of* InfectVax *on a graph G with n vertices, m edges, and neighbourhood diversity w, this instance can be solved in $2^{O(w \log(b+w))} \cdot O(n + m)$ time. Thus,* InfectVax *is FPT when parameterised by neighbourhood diversity and budget.*

Proof. By Lemma 4.2, we need only consider strategies that specify which neighbourhood types to target for inoculation, not specific vertices. At each time-step, we can inoculate up to b vertices drawn from the w

types. The number of ways to choose a multiset of size at most b from w types is $\sum_{i=0}^{b} \binom{w+i-1}{i} = \binom{b+w}{b}$. By Lemma 4.5, there are at most w time-steps, so the total number of type-level strategies is at most $\binom{b+w}{b}^w$. Using the standard inequality $\binom{n}{k} \leq (en/k)^k$ (see, for example, [13]) and symmetry $\binom{b+w}{b} = \binom{b+w}{w}$, we get

$$\binom{b+w}{b}^w \leq \left(\frac{e(b+w)}{\min\{b,w\}}\right)^{w \min\{b,w\}} = 2^{O\left(w \min\{b,w\} \log(b+w)\right)}.$$

Applying Lemma 4.1 with $f(w) = w$, $g(w) = \binom{b+w}{b}$ and $h(n,w) = O(n+m)$, we obtain a runtime of $\binom{b+w}{b}^w \cdot O(n+m) \in 2^{O(w \log(b+w))} \cdot O(n+m)$. ■

4.3 Graphs without long induced paths

A graph is *P_ℓ-free* if it contains no induced path on ℓ vertices. Note that this class is usually called P_k-free, but k is already a variable in FIREFIGHTER. Fomin et al. [11] show that FIREFIGHTER on P_ℓ-free graphs on n vertices is solvable in $\mathcal{O}(n^\ell)$ time. Hunter and Enright [16] follow a similar approach for FIREFIGHTER with costs to inoculate each vertex. Building on these two results, we show that INFECTVAX on P_ℓ-free graphs is also tractable for fixed budgets b and maximum induced path length $\ell - 1$.

Theorem 4.7. *Given an instance (G, r, b, k) of* INFECTVAX *on a P_ℓ-free graph G with n vertices and m edges, this instance can be solved in $\mathcal{O}((n+m)\, n^{b(\ell-1)})$ time.*

Proof. Since G is P_ℓ-free, no open vertex can be infected after time-step $\ell - 1$. Hence we need to only enumerate inoculation strategies of length up to $b(\ell-1)$, of which there are at most $\mathcal{O}(n^{b(\ell-1)})$. For each candidate, we simulate the process and validate the outcome by a search from r in $\mathcal{O}(n+m)$ time to count saved vertices at containment. The runtime of this procedure is $\mathcal{O}((n+m)\, n^{b(\ell-1)})$. ■

Corollary 4.8. *Solving* INFECTVAX *on graphs of maximum path length ℓ is in XP parameterised by ℓ and the budget.*

Since *cographs* and *split graphs* are P_4-free and P_5-free respectively, Theorem 4.7 implies the existence of algorithms for solving INFECTVAX with budget b on cographs and split graphs on n vertices, m edges, that run in $\mathcal{O}((n+m)\, n^{3b})$ and $\mathcal{O}((n+m)\, n^{4b})$ time respectively.

5 Containment

A central problem in graph searching is how to achieve a given objective, such as capturing an intruder or containing a fire, while minimising the resources used at each time-step (see the book [4], for example). In this section, we fix the budget to be $b = 1$ and study containment on two vertex-transitive families of infinite graphs: the n-dimensional Cartesian grid and the n-dimensional strong grid. Since the graphs under consideration are infinite, we do not aim to bound the number of saved vertices, but instead aim to bound the number of time-steps until containment, which bounds the number of contagious vertices. Since we do not have a target k of vertices to save, we let (G, r, b) be the triple consisting of a graph G with root r and positive integer b and consider INFECTVAX on input (G, r, b).

Definition 5.1. A *containment strategy* for an instance of INFECTVAX on input (G, r, b) is a strategy $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_m)$ for some finite integer m, with the property that at least one vertex is open at the start of time-step m, but no open vertex becomes infected during time-step m.

Figure 1 illustrates a containment strategy for the infinite 2-dimensional Cartesian grid: one vertex is inoculated during each of the first four time-steps, the containment strategy has length 5, and containment does not occur until time-step 5.

For an instance of INFECTVAX on input (G, r, b), suppose there is exactly one vertex, v, inoculated at time-step t, but no vertices are inoculated before or after t. The vaccinations stemming from the inoculation of v will reach every vertex in the set $\{u \in V(G) : t + d(v,u) \leq d(r,u)\}$ before the contagion. More generally, let σ be a strategy for INFECTVAX on input (G, r, b). If v is inoculated during time-step t following strategy σ, then the vertices in $\{u \in V(G) : t + d(v,u) \leq d(r,u)\}$ are vaccinated as a direct consequence of the inoculation of v during time-step t. We formally name this set.

Definition 5.2. Let σ be a strategy on input (G, r, b) and v be a vertex inoculated at time-step t. The *transmission set* of v is denoted by $\mathcal{T}(v)$ and defined to be

$$\mathcal{T}(v) = \{u \in V(G) : t + d(v, u) \leq d(r, u)\}.$$

It is important to note that a vertex may be in the transmission set of more than one inoculated vertex. Transmission sets of inoculated vertices will be useful in capturing which vertices in a graph become vaccinated versus infected.

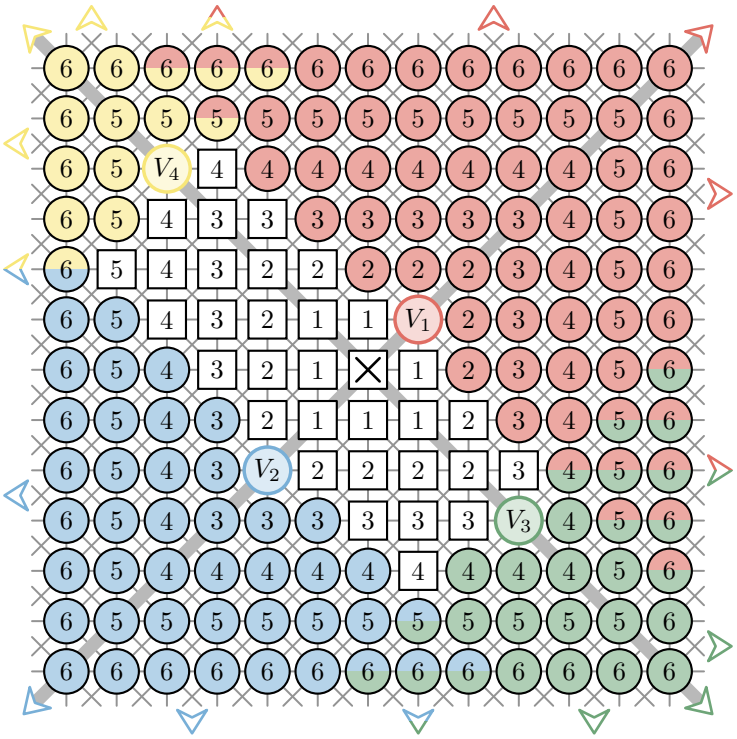


Figure 2: Using budget $b = 1$, a strategy for InfectVax on the infinite 2-dimensional strong grid in which containment occurs at time-step 6.

Figure 2 provides a containment strategy for the infinite 2-dimensional strong grid: one vertex is inoculated during each of the first four time-steps, the containment strategy has length 6, and containment does not occur until time-step 6. The transmission set of each inoculated vertex is indicated by a different colour. Observe from Figure 2 that the contagion spreads to only one vertex during time-step 5. If we had chosen that vertex to be inoculated during time-step 5, then containment would have occurred during time-step 5 instead of time-step 6. However, such a reduction in the number of time-steps to containment comes at the expense of a fifth inoculated vertex. This motivates the next definition.

Definition 5.3. Let $\sigma = (\sigma_1, \ldots, \sigma_m)$ be a containment strategy on input (G, r, b).

1. The *inoculation cost* of σ is $\sum_{i=1}^{m} |\sigma_i|$.
2. The *inoculation number* on input (G, r, b), denoted $\text{inoc}(G, r, b)$, is the minimum inoculation cost, taken over all containment strategies on (G, r, b).
3. The *containment time* on input (G, r, b) is the minimum length of σ, taken over all containment strategies of minimum inoculation cost.

To illustrate the above definitions, consider the containment strategy described by Figure 2. For $b = 1$, the inoculation number is at most 4 and the containment time is at most 6. As a second illustration, in Section 3,

we showed that for an interval graph G and arbitrary vertex r, $\text{inoc}(G, r, 1) \leq 2$ and the containment time is at most 3.

Finally, throughout this section, we will often use the notation $[n]$ to refer to the set of the first n positive integers $\{1, 2, \ldots, n\}$.

5.1 The infinite n-dimensional Cartesian grid

In b-Firefighter, the fire cannot be contained on the infinite 2-dimensional Cartesian grid with budget $b = 1$ [19]; and for $n \geq 3$, the fire cannot be contained on the infinite n-dimensional Cartesian grid with budget $b = 2n-2$ [7]. In stark contrast, we will show that for InfectVax with budget $b = 1$, the contagion can be contained on the infinite n-dimensional Cartesian grid for all n; and we additionally bound the containment time. Let G_n denote the infinite n-dimensional Cartesian grid; that is, the Cartesian product of n infinite two-way paths. For simplicity of notation, we denote vertices in G_n with vector notation. Throughout this section we assume, without loss of generality, that the root of G_n is the origin $\vec{0} = (0, 0, \ldots, 0)$. Observe that for any two vertices $\vec{u} = (u_1, u_2, \ldots, u_n)$ and $\vec{w} = (w_1, w_2, \ldots, w_n)$ in $V(G_n)$, the distance between $\vec{u}$ and $\vec{w}$ is

$$d(\vec{u}, \vec{w}) = \sum_{i=1}^{n} |u_i - w_i|.$$

For $i \in [n]$, let J_i^+ (J_i^-) denote the set of n-tuples in which the ith entry is a positive (negative) integer and all other entries are zero. We define the *axes* of G_n to be $J_1^+, J_2^+, \ldots, J_n^+, J_1^-, J_2^-, \ldots, J_n^-$.

Theorem 5.4. *Let σ be a containment strategy on input $(G_2, \vec{0}, 1)$. The transmission set of each vertex of σ contains vertices from at most one axis of G_2.*

Proof. Let $\vec{u} = (u_1, u_2)$ be a vertex of σ, inoculated at some time-step $t \geq 1$. Without loss of generality, $u_1 > 0$ and $u_1 \geq u_2 \geq 0$. Let $\vec{w} = (0, w_2) \in J_2^+$. Then $t + d(\vec{u}, \vec{w}) = t + u_1 + |u_2 - w_2| > w_2 = d(\vec{0}, \vec{w})$. Thus, $\vec{w} \notin \mathcal{T}(\vec{u})$ and $\mathcal{T}(\vec{u})$ contains no vertices of J_2^+. A similar argument shows $\mathcal{T}(\vec{u})$ contains no vertices of J_1^- or J_2^-. The corresponding argument can be applied to inoculated vertices in other quadrants of G_2. ■

Corollary 5.5 follows immediately as it can be applied to each pair of the $2n$ axes in G_n.

Corollary 5.5. *Let σ be a containment strategy on input $(G_n, \vec{0}, 1)$ for $n \geq 2$. The transmission set of each vertex of σ contains vertices from at most one axis of G_n.*

Corollary 5.5 provides the lower bound for the minimum cost of a containment strategy on $(G_n, \vec{0}, 1)$, which we use in the proof of the next result.

Theorem 5.6. *For $n \geq 2$, $\text{inoc}(G_n, \vec{0}, 1) = 2n$ and the containment time on $(G_n, \vec{0}, 1)$ is at most $n^2 + 1$.*

Proof. Let $\sigma = (\sigma_1, \ldots, \sigma_m)$ be a minimum cost containment strategy on input $(G_n, \vec{0}, 1)$. Since G_n has $2n$ axes and each transmission set contains vertices from at most one axis, the lower bound follows from Corollary 5.5. To prove $\text{inoc}(G_n, \vec{0}, 1) \leq 2n$, we next provide a sequence σ of cost $2n$ and prove it is a containment strategy. For each $i \in [n]$, let $\vec{x}_{2i-1} \in J_i^+$ with i^{th} coordinate $x_i = 2i - 1$ and $\vec{y}_{2i} \in J_i^-$ with i^{th} coordinate $y_i = -2i$. Set $\sigma_{2i-1} = \{\vec{x}_{2i-1}\}$ and $\sigma_{2i} = \{\vec{y}_{2i}\}$ for $i \in \{1, 2, \ldots, n\}$; and set $\sigma_i = \emptyset$ for $i \in \{n+1, \ldots, m\}$. Observe that $d(\vec{x}_{2i-1}, \vec{0}) = 2i - 1$, but since $\vec{x}_{2i-1}$ is inoculated before the contagion spreads during time-step $2i - 1$, $\vec{x}_{2i-1}$ is open at the beginning of time-step $2i - 1$ and will be inoculated. Similarly, $\vec{y}_{2i}$ is open at the beginning of time-step $2i$. Thus, to prove σ is a containment strategy, it suffices to prove there is some finite time-step m during which no open vertex becomes infected. For a contradiction, assume that at every finite time-step, the contagion spreads to an open vertex. Then, for some minimum time-step t, there is a $j \in [n]$ for which

(1) a vertex $\vec{w} = (w_1, \ldots, w_n)$ with $w_j = 2j - 1$ becomes infected at some time-step $t \geq 2j - 1$; or

(2) a vertex $\vec{w} = (w_1, \ldots, w_n)$ with $w_j = -2j$ becomes infected at some time-step $t \geq 2j$.

First, suppose (1) occurs. Observe that

$$t \geq d(\vec{0}, \vec{w}) = \sum_{i=1}^{n} |w_i| = \big(\sum_{i=1}^{n} |w_i|\big) - (2j-1) + (2j-1) = d(\vec{x}_{2j-1}, \vec{w}) + 2j - 1.$$

Therefore, the earliest time-step that $\vec{w}$ can become infected is time-step $d(\vec{x}_{2j-1}, \vec{w}) + 2j - 1$. By definition, open vertices that are adjacent to vaccinated vertices, become vaccinated at the *beginning* of a time-step, whereas open vertices that are adjacent to infected vertices, become infected *later* in a time-step. Thus, $\vec{w}$ will become vaccinated during time-step $d(\vec{x}_{2j-1}, \vec{w}) + 2j - 1$ and (1) cannot occur.

Second, suppose (2) occurs. Observe that

$$t \geq d(\vec{0}, \vec{w}) = \sum_{i=1}^{n} |w_i| = \big(\sum_{i=1}^{n} |w_i|\big) - (2j) + (2j) = d(\vec{y}_{2i}, \vec{w}) + 2j.$$

By a similar argument to (1), $\vec{w}$ will become vaccinated during time-step $d(\vec{y}_{2j}, \vec{w}) + 2j$ and (2) cannot occur.

Therefore, σ is a containment strategy with inoculation cost $2n$ and containment time m. Furthermore, if some vertex $\vec{w} = (w_1, w_2, \ldots, w_n)$ becomes infected during the process, then $-2j < w_j < 2j - 1$ for all $j \in \{1, 2, \ldots, n\}$. Assume $\vec{w}$ becomes infected during time-step $m - 1$. Since the inoculated vertices are all on axes, $m - 1 = d(\vec{0}, \vec{w})$ and

$$m - 1 = d(\vec{0}, \vec{w}) = \sum_{i=1}^{n} |w_i| \leq \sum_{i=1}^{n} |-2i+1| = \sum_{i=1}^{n} (2i-1) = n^2.$$

■

Corollary 5.5 implies that $\text{inoc}(G_n, \vec{0}, b) \geq 2n$ for all $b \geq 1$. Observe that $\text{inoc}(G, r, b) \leq \text{inoc}(G, r, 1)$ for any graph G, fixed root r, and $b > 1$. Corollary 5.7 follows immediately.

Corollary 5.7. *For $b \geq 1$, $\text{inoc}(G_n, \vec{0}, b) = 2n$.*

With respect to the containment time on $(G_n, \vec{0}, 1)$, we conjecture that the bound in Theorem 5.6 is best possible. Since $\text{inoc}(G_n, \vec{0}, b) = 2n$ for all $b \geq 1$, the containment time on $(G_n, \vec{0}, b)$ is bounded above by the containment time for $(G_n, \vec{0}, 1)$.

Question 5.8. *For $b \geq 1$, what is the containment time for $(G_n, \vec{0}, b)$?*

5.2 The infinite n-dimensional strong grid

The *strong product* of two graphs H and H', written $H \boxtimes H'$, has vertex set $V(H) \times V(H')$. Vertices (a, b) and (c, d) in $H \boxtimes H'$ are adjacent if (1) $a = c \in V(H)$ and $(b, d) \in E(H')$, or (2) $(a, c) \in E(H)$ and $b = d \in V(H')$, or (3) $(a, c) \in E(H)$ and $(b, d) \in E(H')$. Let H_n denote the infinite n-dimensional strong grid; that is, the strong product of n infinite two-way paths. We continue to use vector notation, and as in Section 5.1, we assume, without loss of generality, that the root of H_n is the origin $\vec{0}$. We again focus firstly on minimising budget and secondly in minimising the number of inoculations. That is, we show that the contagion can be contained with a budget of only $b = 1$ and then determine the minimum cost of a containment strategy for $(H_n, \vec{0}, 1)$. Note that for any $\vec{u} = (u_1, u_2, \ldots, u_n)$ and $\vec{w} = (w_1, w_2, \ldots, w_n)$ in $V(H_n)$, the distance between $\vec{u}$ and $\vec{w}$ is

$$d(\vec{u}, \vec{w}) = \max_{i \in [n]} |u_i - w_i|.$$

Let A_n denote the set of all vectors of length n whose entries are from $\{-1, 1\}$.

The Hadamard product is a binary operation that takes two vectors of the same length and returns a vector of the multiplied corresponding elements. For example, the Hadamard product of (a, b, c) and (x, y, z) is $(a, b, c) \circ (x, y, z) = (ax, by, cz)$. Note that we are using vector notation for convenience, but $\vec{a} \circ \vec{x}$ and $x \cdot \vec{a}$ represent *vertices* in the graph.

For each $\vec{a} \in A_n$, define the *orthant of* $\vec{a}$ to be[1]

$$\mathcal{O}_{\vec{a}} = \{\vec{a} \circ \vec{x} \ : \ x_i \in \mathbb{N} \cup \{0\} \ \forall i \in [n] \text{ and } \vec{x} \neq \vec{0}\}.$$

Thus, $\vec{a}$ dictates the signs of the coordinates of the vertices in $\mathcal{O}_{\vec{a}}$. Define the *set of all orthants of* H_n to be

$$\mathcal{O} = \bigcup_{\vec{a} \in A_n} \mathcal{O}_{\vec{a}}$$

and note that $\mathcal{O} = V(H_n) \backslash \{\vec{0}\}$. For each $\vec{a} \in A_n$, the *diagonal orthant of* $\vec{a}$ is the set $\{x \cdot \vec{a} \ : \ x \in \mathbb{N}\}$.

To illustrate the above definitions, consider the infinite 2-dimensional strong grid H_2 and observe:

- $A_2 = \{(1,1), (-1,-1), (-1,1), (1,-1)\}$;
- $\mathcal{O}_{(1,1)} = \{(u,v) \in \mathbb{Z} : u, v \geq 0 \text{ and at least one of } u, v \text{ is non-zero}\}$;
- $\mathcal{O}_{(-1,-1)} = \{(u,v) \in \mathbb{Z} : u, v \leq 0 \text{ and at least one of } u, v \text{ is non-zero}\}$;
- $\mathcal{O}_{(-1,1)} = \{(u,v) \in \mathbb{Z} : u \leq 0, v \geq 0, \text{ and at least one of } u, v \text{ is non-zero}\}$;
- $\mathcal{O}_{(1,-1)} = \{(u,v) \in \mathbb{Z} : u \geq 0, v \leq 0, \text{ and at least one of } u, v \text{ is non-zero}\}$.

Note that vertices of the form $(u,0)$ or $(0,v)$ for $u, v \neq 0$ are in the orthant of two different vertices of A_2. For example, for $u > 0$, $(u,0)$ is in both $\mathcal{O}_{(1,1)}$ and $\mathcal{O}_{(1,-1)}$. For $n = 2$, the diagonal orthants are highlighted with thick grey lines in Figure 2. Figure 2 also illustrates how the minimum cost of a containment strategy, that is, the inoculation cost, for $(H_n, \vec{0}, 1)$ is at most four.

The following lemma will help to determine the minimum cost of a containment strategy for the infinite n-dimensional strong grid in an analogous fashion to how Theorem 5.4 helped to prove the minimum cost of a containment strategy for the infinite n-dimensional Cartesian grid in Section 5.1.

Lemma 5.9. *For $n \geq 2$, consider a containment strategy σ on input $(H_n, \vec{0}, 1)$. Each transmission set contains vertices from at most one diagonal orthant of H_n.*

Proof. Let σ be a containment strategy on input $(H_n, \vec{0}, 1)$. Denote by $D_{\vec{a}}$ and $D_{\vec{b}}$, two distinct diagonal orthants of H_n. As a contradiction, suppose $\vec{u} = (u_1, u_2, \ldots, u_n)$ is inoculated during time-step t and $\mathcal{T}(\vec{u})$ contains both vertices $\vec{x} = (a_1 x, a_2 x, \ldots, a_n x) \in D_{\vec{a}}$ and $\vec{y} = (b_1 y, b_2 y, \ldots, b_n y) \in D_{\vec{b}}$ for some positive integers x and y. Without loss of generality, assume $x < y$.

Since $\vec{y} \in \mathcal{T}(\vec{u})$ and $\vec{u}$ is inoculated during time-step t, by Definition 5.2

$$t + \max_{i \in [n]} |u_i - b_i y| = t + d(\vec{u}, \vec{y}) \leq d(\vec{0}, \vec{y}) = y.$$

Since $t \geq 1$, it follows that $\max_{i \in [n]} |u_i - b_i y| < y$. We similarly deduce that $\max_{j \in [n]} |u_j - a_j x| < x$. As $D_{\vec{a}} \neq D_{\vec{b}}$, there must be some $i \in [n]$ such that $b_i \neq a_i$. Without loss of generality, suppose $b_1 \neq a_1$. Then $b_1 = -a_1$ as $b_1, a_1 \in \{-1, 1\}$. It follows that

$$\text{(a)} \ \ |u_1 - a_1 x| < x \quad \text{and} \quad \text{(b)} \ \ |u_1 + a_1 y| < y. \tag{1}$$

If $a_1 = -1$ then Inequality (1)(a) simplifies to $|u_1 + x| < x$. As $x > 0$, this implies $u_1 < 0$, which contradicts Inequality (1)(b). Similarly, if $a_1 = 1$ then Inequality (1)(b) simplifies to $|u_1 + y| < y$. As $y > 0$, this implies $u_1 < 0$, which contradicts Inequality (1)(a). ■

As there are 2^n distinct diagonal orthants, Lemma 5.9 implies $\text{inoc}(G) \geq 2^n$.

Recall that A_n denotes the set of length n vectors, whose coordinates are in $\{-1, 1\}$. As there is a bijection between the vectors of A_n and the set of bit-strings of length n, we arbitrarily label the vectors in A_n as $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_{2^n}$. In the next result, we present a sequence σ and show that it bounds from above, the coordinates of any infected vertex. We then prove that σ is a containment strategy of length 2^n for $(H_n, \vec{0}, 1)$ and provide an upper bound for the containment time.

[1] A similar definition is used in matroid theory. In particular, $\vec{v}$ *conforms* to $\vec{a}$ if each non-zero coordinate of $\vec{v}$ has the same sign as the corresponding coordinate in $\vec{a}$. Then $\mathcal{O}_{\vec{a}}$ is defined to be the set of all non-zero (integer-valued) vectors that conform to $\vec{a}$.

Lemma 5.10. *Let $A_n = \{\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_{2^n}\}$ and $\sigma = (\sigma_1, \ldots, \sigma_m)$ where*

$$\sigma_i = \begin{cases} \{\vec{a}_i \cdot i\} & \text{if } i \in \{1, \ldots, 2^n\}, \\ \emptyset & \text{if } i \in \{2^n + 1, \ldots, m\}. \end{cases}$$

Let $\vec{w} \neq \vec{0}$ be a vertex infected by the end of time-step m. Then for some $x \in \{1, 2, \ldots, 2^n\}$, $\vec{w} \in \mathcal{O}_{\vec{a}_x}$. Let $\vec{a}_y \in A_n$ such that $\vec{a}_y$ differs from $\vec{a}_x$ only in coordinate j for some $j \in [n]$. Then $|w_i| < x + y$ for all $i \neq j$.

Proof. We express $\vec{w}$ as $\vec{w} = (a_1 w_1, a_2 w_2, \ldots, a_n w_n)$ where each w_i is a nonnegative integer (at least one being positive) and each $a_i \in \{-1, 1\}$. Then $\vec{a}_x = (a_1, a_2, \ldots, a_n) \in A_n$ and $\vec{w} \in \mathcal{O}_{\vec{a}_x}$. Let $\vec{a}_y = (b_1, b_2, \ldots, b_n)$ where $\vec{a}_y$ differs from $\vec{a}_x$ only in coordinate j. Without loss of generality, let $j = 1$; and in particular, let $a_1 = 1$ and $b_1 = -1$. Following σ, note that vertices $\vec{a}_x \cdot x$ and $\vec{a}_y \cdot y$ are inoculated at time-steps x and y, respectively. Without loss of generality, suppose $x > y$.

We aim to prove $w_i < x + y$ for $i \neq j$. For a contradiction, we assume $w_i \geq x + y$ for some $i \neq 1$. However, if such a vertex $\vec{w}$ is infected at time-step t, then there is an earlier time-step $t' < t$ in which $\vec{v}$ is infected, where $\vec{v} = (a_1 v_1, \ldots, a_n v_n)$ with $v_1 \geq 0$, $0 < v_j \leq x + y$ for $j \neq 1$, and $v_j = x + y$ for some $j \in [n]$.

Then $\vec{v} = (b_1 v_1', b_2 v_2', \ldots, b_n v_n')$ where $b_1 = -a_1$, $v_1' = -v_1$; and for $i > 1$, $b_i = a_i$ and $v_i' = v_i$. Since $\vec{v} \notin \mathcal{T}(\vec{a}_y \cdot y)$,

$$x + y = \max_{i \in [n]} v_i = d(\vec{0}, \vec{v}) < y + d(\vec{a}_y \cdot y, \vec{v}) = y + \max_{i \in [n]} |b_i y - b_i v_i'| = y + \max_{i \in [n]} |y - v_i'|$$

$$\iff x < \max_{i \in [n]} \big\{ y + v_1, |y - v_2|, |y - v_3|, \ldots, |y - v_n| \big\}.$$

Case 1: Suppose $\max_{i \in [n]} |y - v_i'| = |y - v_r|$ for some $r > 1$. Then $x < |y - v_r|$. If $v_r \geq y$ then $v_r - y > x \iff v_r > x + y$, a contradiction. Thus, $y > v_r$ and furthermore,

$$\max_{i \in [n]} |y - v_i'| = |y - v_r| \geq y + v_1 \iff y - v_r \geq y + v_1 \iff v_1 + v_r \leq 0,$$

which implies $v_1 = v_r = 0$. But then $x < |y - v_r| \iff x < y$, which contradicts $x > y$. Thus, Case 1 cannot occur.

Case 2: Suppose $\max_{i \in [n]} |y - v_i'| = y + v_1$. Then $v_1 > x - y$. Since Case 1 does not occur, $y + v_1 > |y - v_i|$ for all $i > 1$. We consider $|y - v_i|$ for each $i > 1$.

- If $v_i \leq y$ then $v_i \leq y < x + y$.
- If $v_i > y$ then $y + v_1 > |y - v_i| = v_i - y \iff 2y > v_i - v_1$.
  - If $v_i \geq v_1$ then $0 \leq v_i - v_1 < 2y < x + y$, which implies $v_i < x + y$.
  - If $v_i < v_1$ then $v_i < v_1 \leq x + y$.

  In all situations, we find $v_i < x + y$ for all $i > 1$, a contradiction.

For Case 2, there is no situation in which $v_i = x + y$ for any $i > 1$.

Thus, if $\vec{w} \neq \vec{0}$ is infected by the end of time-step m, then $w_i < x + y$ for all $i \neq j$. A similar argument shows that $w_i > -(x + y)$ for all $i \neq j$. Thus, $|w_i| < x + y$ for all $i \neq j$. ■

We next introduce a parameter that will provide an upper bound for the containment time for $(H_n, \vec{0}, 1)$. Let G be a graph and f be an ordering of the vertices of G. The *strength of* f, denoted $\mathrm{str}_f(G)$, is

$$\mathrm{str}_f(G) = \max \big| \{ f(u) + f(v) \ : \ uv \in E(G) \} \big| \tag{2}$$

and the *strength* of G, denoted $\mathrm{str}(G)$, is

$$str(G) = \min \big| \{ \mathrm{str}_f(G) \ : \ f \text{ is a linear ordering of } G \} \big|. \tag{3}$$

The parameters of (2) and (3) were first defined in [17], motivated by super edge-magic labelings and the super magic strength of a graph. The strength of a graph is a generalisation of the concept of super magic strength. See [2, 17] for more on super edge-magic labelings and super magic strength. Table 1 details exact values and best known upper bounds for $\mathrm{str}(Q_n)$. We make use of the results in Table 1 to bound the containment time for $(H_n, \vec{0}, 1)$.

Theorem 5.11. *For $n \geq 2$, $\text{inoc}(H_n, \vec{0}, 1) = 2^n$ and the containment time is at most $str(Q_n)$.*

Proof. That $\text{inoc}(H_n, \vec{0}, 1) \geq 2^n$ follows from Lemma 5.9. To prove $\text{inoc}(H_n, \vec{0}, 1) \leq 2^n$ and provide an upper bound on the containment time, we first define a sequence $\sigma = (\sigma_1, \ldots, \sigma_m)$: let $\sigma_t = \{\vec{a}_t \cdot t\}$ for $t \in \{1, \ldots, 2^n\}$ and $\sigma_t = \emptyset$ for $t \in \{2^n + 1, \ldots, m\}$.

Recall that the elements of A_n were given an arbitrary ordering $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_{2^n}$. We apply Lemma 5.10 to each pair of elements in A_n that differ in exactly one coordinate. As a result, the largest coordinate of any infected vertex is at most

$$\max\big|\{i + j - 1 : i, j \in \{1, 2, \ldots, 2^n\} \text{ and, } \vec{a}_i \text{ and } \vec{a}_j \text{ differ in exactly one coordinate}\}\big|. \tag{4}$$

If m is larger than the quantity in (4), the contagion is contained, and so σ is a containment strategy for $(H_n, \vec{0}, 1)$. Furthermore, the quantity in (4) provides us with an upper bound for the containment time of $(H_n, \vec{0}, 1)$.

Each element of A_n corresponds to a distinct vertex in the hypercube Q_n. So $\vec{a}_i$ and $\vec{a}_j$ differ in exactly one coordinate if and only if corresponding vertices x_i and x_j in Q_n are adjacent. Given an arbitrary ordering of the elements of A_n, any bijection $f : \{x_1, x_2, \ldots, x_{2^n}\} \to \{1, 2, \ldots, 2^n\}$ will correspond to the following bound on the containment time:

$$\max\big|\{f(u) + f(v) : uv \in E(Q_n)\}\big| = \max\big|\{f(u) + f(v) : uv \in E(Q_n)\}\big| = str_f(Q_n).$$

Thus, the containment time for $(H_n, \vec{0}, 1)$ is at most

$$\text{str}(Q_n) = \min\big|\{\text{str}_f(Q_n) : f \text{ is a linear ordering of } Q_n\}\big|.$$

■

The containment time for $(H_n, \vec{0}, 1)$ is therefore bounded by the values given in Table 1.

n	$\text{str}(Q_n)$	comment
1	$= 3$	due to [17]
2	$= 6$	due to [17]
3	$= 11$	due to [12]
4	$= 21$	due to [12]
5	$= 40$	due to [12]
6	≤ 78	due to [15]
7	≤ 152	due to [15]

n	$\text{str}(Q_n)$	comment
8	≤ 300	due to [15]
9	≤ 591	due to [15]
10	≤ 1173	due to [15]
11	≤ 2323	due to [15]
12	≤ 4623	due to [15]
13	≤ 9181	due to [15]
14+	$\leq 2^n + 2^{n-3} + 28$	due to [15]

Table 1: Known values and best current upper bounds for $\text{str}(Q_n)$.

Similar to the n-dimensional Cartesian grid, the inoculation number of H_n is 2^n for all $b \geq 1$.

Corollary 5.12. *For $b \geq 1$, $\text{inoc}(H_n, \vec{0}, b) = 2^n$.*

The lower bound in Corollary 5.12 follows from Lemma 5.9 and the upper bound follows because we have $\text{inoc}(G, r, b) \leq \text{inoc}(G, r, 1)$ for any graph G, fixed root r, and $b > 1$. The containment time on $(H_n, \vec{0}, b)$ is bounded above by the containment time on $(H_n, \vec{0}, 1)$, but the actual values remain unknown.

Question 5.13. *For $b \geq 1$, what is the containment time for $(H_n, \vec{0}, b)$?*

5.3 The inoculation number of subgraphs

We conclude the paper with a result about subgraphs. A natural question is: *if H is an induced subgraph of G, then is the inoculation number of H at most that of G?* In general, the answer is no, even when the subgraph is connected or isometric as shown by Theorem 5.14 below.

Let G^* be the graph shown on the left-side of Figure 3 where the ellipses indicate paths of infinite length. The subgraphs induced by the deletion of center c from G^* are labelled B_1, B_2, and B_3; and observe that

$B_1 \cong B_2 \cong B_3$. Let H^* be the subgraph illustrated on the right-side of Figure 3 and observe it is the subgraph of G^*, induced by the deletion of three particular vertices.

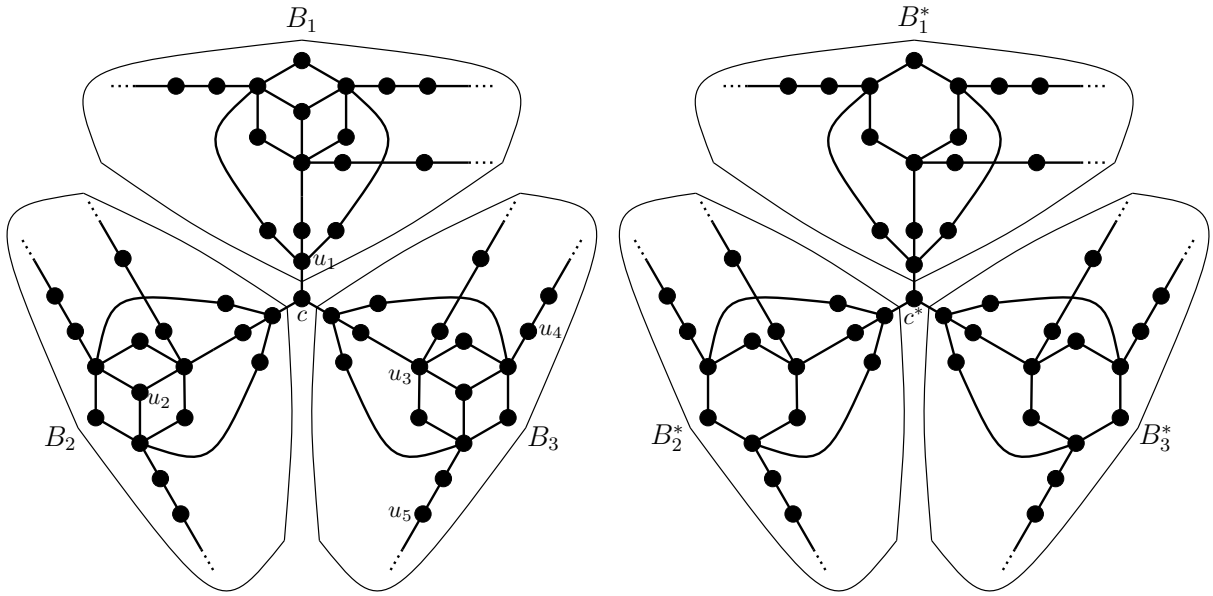


Figure 3: Graph G^* (left) and induced subgraph H^* (right).

Theorem 5.14. *For graph G^* and induced subgraph H^* described by Figure 3,* $\text{inoc}(G^*, r, 1) < \text{inoc}(H^*, r, 1)$ *for every vertex r that appears in both G^* and H^*.*

Proof. We first prove that $\text{inoc}(G^*, r, 1) \leq 5$ by considering two cases: whether the root r is c or not. First, suppose the root is c. If u_i is inoculated at time-step i for $i \in [5]$ as illustrated in Figure 3, the reader can verify the contagion will be contained within 5 time-steps. Second, suppose the root r is not c. Without loss of generality, r is in B_1. If r is a degree-two vertex on an infinite path, the contagion can be contained within two time-steps. Otherwise, we inoculate c during time-step 1 to prevent the contagion from spreading to B_2 or B_3. For each of the next 3 time-steps, we inoculate a degree-two vertex on an infinite path that is adjacent to an infected vertex. In doing so, the contagion will be contained within four time-steps. Thus, $\text{inoc}(G^*, r, 1) \leq 5$.

We next prove $\text{inoc}(H^*, r, 1) \geq 6$. It suffices to show that if the root is c^*, then we will require at least 6 inoculations for containment. Observe that at most one neighbour of c^* can be inoculated during time-step 1, so the contagion will infect vertices in at least two of B_1^*, B_2^*, B_3^*. Without loss of generality, a vertex of B_1^* is inoculated during time-step 1. The contagion then spreads from c^* to its neighbours in B_2^* and B_3^*. Without loss of generality, the vertex inoculated during time-step 2 is not in B_3^*. Then, no vertex of B_3^* is inoculated during time-steps 1 and 2. Containing the contagion on B_3^* requires the inoculation of three additional inoculated vertices, which equates to three additional time-steps. We must, however, also consider B_2^*, in which at most one vertex is inoculated during time-step 2. Note that at least one additional vertex must be inoculated in B_2^* in order to contain the virus on B_2^*. Thus, since at least 6 vertices must be inoculated in order to contain the virus on H^*, $\text{inoc}(G^*, r, 1) \leq 5 < 6 \leq \text{inoc}(H^*, r, 1)$. ■

Since H^* is an induced subgraph of G^*, it is not true that $\text{inoc}(H, r, b) \leq \text{inoc}(G, r, b)$ for every induced subgraph H of a graph G. We note that the inequality remains untrue even if the induced subgraph is required to be connected since H^* is a connected induced subgraph of G^*. We also note that the inequality remains untrue even if the subgraph is required to be isometric since H^* is an isometric subgraph of G^*. Subgraph H^* is isometric because, for every pair of vertices $u, v \in V(H^*)$, the distance between u and v in G^* is the same as the distance between u and v in H^*.

Acknowledgements

D. Pearson acknowledges research support from the Graph Searching in Atlantic Canada (GSAC) collaborative group, the Atlantic Association for Research in Mathematical Sciences (AARMS), and Mount Allison University. Melissa A. Huggan acknowledges research support from NSERC (grant application 2023-03395 and DGECR-2023-00190). M.E. Messinger acknowledges research support from NSERC (grant application 2018-04059).

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